Human-AI Interaction Safety Mechanism Analysis - Research and Coding Tasks Mini Project

The project conducted by the Garrick Institute of Risk Sciences aims to:

1. Examine the mechanisms by which humans interact with autonomous systems, including both workflow-level interaction and cognitive-flow processes.

2. Identify the novel characteristics and features of modern AI / autonomous systems in comparison with conventional automation used in high-risk industries.

E.g., healthcare, power generation, transportation, oil and gas, maritime, manufacturing. These features may include, but are not limited to:

(1) Non-Deterministic, Data-Driven Decision Making. Why it matters: outputs vary with data and model states → uncertainty + unpredictable failure modes.

Industries where this feature is widely present and heavily discussed:

- Healthcare: AI diagnosis, radiology, triage algorithms
- Autonomous Driving & Transportation: perception, prediction, trajectory planning
- Aviation: predictive maintenance, flight-path optimization
- Energy & Power Systems: reactor anomaly detection, grid forecasting

(2) Opacity and Lack of Explainability ("Black-Box Behavior"). Why it matters: operators cannot see internal reasoning → trust, regulation, and safety assurance are difficult.

Industries where opacity is a major concern:

- Nuclear Power & Oil/Gas: safety-critical decision support systems
- Healthcare: diagnostic AI, treatment recommendation systems
  Aviation: pilot decision support, autonomous co-pilots
  Autonomous Vehicles: deep perception and end-to-end driving models

(3) Context-Aware and Adaptive Behavior (Dynamic Response to Environment) Why it matters: adaptability improves performance but introduces variability + unpredictability.

Industries where context-aware or environment-adaptive AI is essential:

- Autonomous Driving: reacting to road, weather, traffic, pedestrian behavior
- Maritime Autonomous Navigation: collision avoidance, route adaptation
- Aviation & Drones: adaptive flight planning, turbulence response
- Smart Manufacturing: robots adjusting to variability in objects or human workers
- Healthcare: personalized treatment suggestions, patient-state monitoring

3. Identify the types of human performance degradation that may arise from these new AI features, in contrast with traditional systems. Examples include:

- Over-trust or under-trust in AI
- Automation bias and confirmation bias
- Mis-prioritization of tasks
- Reduced situational awareness
- Mode confusion
- Cognitive overload from complex interface navigation
- Skill degradation over time due to reliance on AI
- Misinterpretation of AI-generated recommendations

Additionally, analyze how specific AI features causally affect human performance. For example, how opacity impacts trust calibration, how adaptive behavior affects predictability, or how autonomous decision suggestions influence operator workload.

4. Identify the measurable metrics for evaluating AI features that may degrade human performance during work, tasks, or operational activities, and explain how to use them . Examples of metrics might be used in current studies include:

- Trust calibration indices
- Situation awareness scales (SAGAT, SART)
- Cognitive workload measures (NASA-TLX, EEG-based metrics)
- Error rate and error recovery time
- Latency in decision making
- Dependence or automation reliance indices
- Transparency and explainability scores

5. Review current research to determine whether real, semi-real, or simulated human-AI interaction platforms or frameworks exist.

It may involve experimental testbeds, software tools, or theoretical approaches for modeling human–autonomous-system interaction. For each platform or framework, document:

- Appearance, structure, and background (country, institution, development history)
- Functions and capabilities
- Operational workflow (input → process → output)
- Topics solved to date and remaining challenges
- Future development plans

Tasks to Be Completed

Task 1 — Literature Collection (Research Skill)

Search for relevant academic publications using Google Scholar. Consider:

- Publication years (preferably within the past 5 years)
- Citation counts
- Journal impact factors
- Industry domains (healthcare, oil and gas, nuclear power, transportation, maritime, manufacturing, education, psychology)

Suggested search keywords include:
Human-AI teaming, Human-AI interaction, Human-autonomous systems interaction, Human-autonomous collaboration, trust in AI, human bias toward AI, human error, human factors, safety, human performance, simulation platform, simulation tool, plus any domain-specific keywords.
You may generate additional keywords as needed. Combining keywords allows for more targeted literature searches.

Expected outcome:
Identify at least **10 relevant academic articles**. Determine relevance based on the project description. Three example articles are provided for reference.

Task 2 — Literature Review (Research Skill)

Based on the five research goals:

1. Extract relevant information from each article (either direct excerpts or summarized findings) and document them in a Word file with proper citations.
2. Summarize your synthesized insights across the reviewed literature.

Expected outcome:
Use the provided Word template. Each article should be summarized as a coherent, interpretable "story" with clear relevance to the project goals.

Task 3 — Evidence Extraction (Coding + Linguistic Data Analysis)

Using the collected literature from Task 1, identify evidence supporting your findings or conclusions. This is part of a long-term plan to build a large-scale database (potentially hundreds of sources) quantifying:

- AI features
- Human performance degradation types
- Causal links between them
- Frequency of occurrence across literature

You are expected to be capable of two technical approaches:

(1) API-Based Solution

Develop code to transmit source texts into LLM tools. You must consider:

- Reasoning accuracy across LLM models
- Time and computational efficiency

(2) NLP Pipeline Solution

You need to understand how to perform:

1. Text chunking
2. Vectorization
3. Embedding
4. Cosine similarity estimation

(3) Prompt Engineering

For both methods above, you must know how to design prompts:

- Key design principles
- How to extract relevant excerpts corresponding to the five major project goals
- How to request and evaluate LLM justifications explaining *why* certain excerpts are relevant

(4) Validation

You must develop code to validate that extracted excerpts **100% accurately** exist in the original data sources.

Task 4 Model Construction and Quantification (Led by GIRS Researchers)

Based on the outputs from Tasks 1–3, we will conduct causal modeling and quantitative analysis to build:

- Human-AI interaction workflow and cognitive-flow models
- Causal models explaining how AI features influence human performance
- A simulation environment for modeling human-AI interactions
- Quantitative results estimating human- and AI-error probabilities across scenarios

Students who perform well in earlier tasks may join this advanced modeling work.

Participation Options for Students

Option A: Task 1 + Task 2 — Research Track
Ideal for students interested in gaining a deep understanding of current research on human-AI interaction. You will develop comprehensive domain knowledge to support future modeling work and industry applications.

Option B: Task 1 + Task 3 — Coding Track
Ideal for students interested in NLP, LLM applications, and software development. You will work closely with GIRS collaborators who develop NLP systems. Your work may be generalized and used in future projects requiring linguistic data analysis.